

Evaluation of Medical Treatment Recommendation Accuracy in an AI-Assisted Clinical Decision-Support System: A 50-Question Benchmark Study

Author: Dominique Ball

Corresponding Author Email: Domhasaball@gmail.com

Article Type: Viewpoint / Tutorial

Abstract

Background

Clinical decision-support systems and artificial intelligence (AI)-assisted tools are increasingly being evaluated for their ability to recognize clinical presentations and generate appropriate treatment recommendations. Accurate identification of emergency medical conditions is particularly important because errors in recognition or treatment recommendations may have significant consequences.

Objective

This study evaluated the accuracy of an AI-assisted clinical decision-support software system in identifying appropriate medical treatments across a standardized set of emergency and medical scenarios.

Methods

A benchmark consisting of 50 clinical questions was developed across eight medical conditions representing respiratory, cardiovascular, allergic, gastrointestinal, and abdominal presentations. The conditions evaluated were bronchospasm/asthma, spontaneous pneumothorax, respiratory

failure, impending cardiac arrest, severe allergic reaction/anaphylaxis, appendicitis, gastrointestinal bleeding, and gallstones/cholecystitis. Five questions were evaluated for each condition. The software's responses were graded for medical treatment accuracy.

Results

The software achieved a treatment accuracy of **100% across all 50 questions**. No treatment recommendation was graded as incorrect in the evaluated benchmark. The system therefore achieved a 50/50 correct treatment-recommendation score, corresponding to an observed accuracy of 100%.

Conclusion

Within the conditions and questions included in this benchmark, the evaluated software demonstrated complete agreement with the predetermined treatment-answer criteria. These findings support further investigation of the system as a potential clinical decision-support tool. However, the results represent performance on a limited 50-question benchmark and should not be interpreted as evidence of clinical safety, efficacy, or readiness for independent use in patient care. Larger, independently developed, clinically representative, and prospectively validated studies are required.

Keywords: artificial intelligence, clinical decision support, treatment recommendation, medical AI, emergency medicine, diagnostic accuracy, treatment accuracy, benchmark evaluation

1. Introduction

Artificial intelligence and computer-assisted clinical decision-support systems have the potential to assist healthcare professionals in recognizing medical conditions and selecting appropriate interventions. In emergency and prehospital medicine, rapid recognition of clinical patterns and appropriate treatment selection are particularly important because clinical deterioration may occur rapidly.

Respiratory emergencies such as asthma/bronchospasm, spontaneous pneumothorax, and respiratory failure require recognition of characteristic clinical findings and prompt selection of appropriate interventions. Similarly, cardiovascular emergencies, anaphylaxis, gastrointestinal hemorrhage, appendicitis, and biliary disease require clinicians to distinguish potentially serious presentations and determine appropriate management.

Despite increasing interest in AI-assisted medical decision support, demonstrating performance requires systematic evaluation against predefined clinical criteria. A system that performs well on a controlled benchmark may still perform differently when confronted with more complex patients, incomplete information, atypical presentations, comorbidities, medication interactions, or situations outside the benchmark.

The present study was designed as an initial benchmark evaluation of treatment recommendation accuracy. The objective was to determine whether the evaluated software could provide medically appropriate treatment recommendations across a predefined set of 50 clinical questions.

The primary outcome was the proportion of questions for which the software's treatment recommendation was graded as medically accurate according to the predetermined evaluation criteria.

2. Materials and Methods

2.1 Study Design

This investigation was conducted as a structured benchmark evaluation of an AI-assisted clinical decision-support software system. The evaluation consisted of 50 clinical questions distributed across eight medical conditions.

Five questions were evaluated for each condition.

2.2 Clinical Conditions

The benchmark included the following conditions:

1. **Bronchospasm/Asthma**
2. **Spontaneous Pneumothorax**
3. **Respiratory Failure**
4. **Impending Cardiac Arrest**
5. **Severe Allergic Reaction/Anaphylaxis**
6. **Appendicitis**
7. **Gastrointestinal Bleeding**
8. **Gallstones/Cholecystitis**

These conditions were selected to represent a range of respiratory, cardiovascular, allergic, gastrointestinal, and abdominal clinical presentations.

2.3 Clinical Presentation Domains

The questions incorporated characteristic clinical findings associated with the target conditions.

For bronchospasm/asthma, the benchmark included findings such as prolonged expiration and bilateral wheezing.

For spontaneous pneumothorax, scenarios included sudden shortness of breath and chest pain with diminished breath sounds on the affected side.

Respiratory failure scenarios incorporated findings such as altered mental status, inability to speak in complete sentences, and respiratory arrest requiring positive-pressure ventilation.

Cardiovascular scenarios included symptoms associated with impending cardiac deterioration, including dyspnea, unusual fatigue, chest discomfort, nausea/vomiting, and back pain.

Anaphylaxis scenarios incorporated findings including acute respiratory difficulty, chest tightness, swelling involving the tongue, face, or neck, and urticaria.

Abdominal and gastrointestinal scenarios included characteristic findings of appendicitis, gastrointestinal hemorrhage, and gallbladder disease. These included right-lower-quadrant abdominal pain, melena, hematochezia, hematemesis, and right-upper-quadrant or epigastric pain with characteristic radiation.

2.4 Evaluation Procedure

Each clinical condition was represented by five questions, producing a total of 50 evaluated questions.

The software was provided the clinical questions and its treatment recommendations were evaluated against the predetermined medical answer criteria.

The primary evaluation metric was treatment recommendation accuracy.

A response was classified as correct when the treatment recommendation satisfied the predefined medical-treatment criteria for the corresponding question.

2.5 Outcome Measure

The primary outcome was calculated as:

$$\text{Treatment Accuracy} = \frac{\text{Number of Correct Treatment Recommendations}}{\text{Total Number of Questions}} \times 100$$
$$\text{\texttt{\text{Treatment Accuracy}}} = \frac{\text{\texttt{\text{Number of Correct Treatment Recommendations}}}}{\text{\texttt{\text{Total Number of Questions}}}} \times 100$$

The maximum possible score was 50 correct treatment recommendations out of 50 questions.

3. Results

3.1 Overall Performance

The software achieved a **100% treatment recommendation accuracy rate** across the 50-question benchmark.

Metric	Result
--------	--------

Total questions 50

Questions graded correct 50

Questions graded incorrect 0

Overall treatment accuracy **100%**

The observed accuracy was therefore:

$$50 \div 50 \times 100 = 100\% \quad \frac{50}{50} \times 100 = 100\%$$

No incorrect treatment recommendations were identified within the evaluated benchmark.

3.2 Performance by Clinical Category

Five questions were evaluated for each of the eight clinical conditions.

Clinical condition	Questions	Correct	Accuracy
Bronchospasm/Asthma	5	5	100%
Spontaneous Pneumothorax	5	5	100%
Respiratory Failure	5	5	100%
Impending Cardiac Arrest	5	5	100%
Severe Allergic Reaction/Anaphylaxis	5	5	100%
Appendicitis	5	5	100%
Gastrointestinal Bleeding	5	5	100%

Gallstones/Cholecystitis	5	5	100%
Total	50	50	100%

The system therefore maintained the same observed accuracy across every clinical category represented in the benchmark.

4. Discussion

The principal finding of this benchmark evaluation was that the evaluated software achieved **100% treatment recommendation accuracy across all 50 questions**. Every question included in the evaluation received a treatment recommendation that met the predetermined grading criteria.

The finding is notable because the benchmark included multiple clinical domains rather than a single disease category. Respiratory emergencies, cardiovascular deterioration, anaphylaxis, gastrointestinal bleeding, appendicitis, and biliary disease present substantially different clinical management considerations. The software nevertheless achieved complete observed accuracy within the tested question set.

However, the interpretation of a 100% benchmark score requires caution. A perfect score on 50 questions does not establish that the software will achieve 100% accuracy in real-world clinical practice. The benchmark represents a finite sample of possible clinical presentations, and the

performance estimate may change substantially with a larger or independently constructed question set.

In addition, clinical cases frequently contain ambiguity, incomplete information, conflicting findings, multiple simultaneous diagnoses, medication contraindications, allergies, age-related considerations, pregnancy, comorbid disease, and rapidly changing patient status. These factors may not be adequately represented by a controlled benchmark.

The distinction between **benchmark accuracy** and **clinical effectiveness** is therefore important. The current results demonstrate that the software successfully answered the questions included in this evaluation according to the specified grading criteria. They do not independently establish patient outcomes, clinical safety, diagnostic performance, or superiority over qualified healthcare professionals.

Another important consideration is potential benchmark bias. If the questions, expected answers, or evaluation criteria were known to the system developer or closely reflected the information used to develop the software, performance could overestimate generalization to previously unseen cases. Future studies should therefore use independently constructed test sets and, where possible, blinded external evaluation.

5. Limitations

Several limitations should be considered.

First, the sample size was limited to 50 questions. Although the system achieved perfect observed accuracy, a relatively small benchmark cannot establish reliable performance across the full range of clinical scenarios.

Second, only eight clinical conditions were evaluated. Many important emergency and medical conditions were not included.

Third, the questions may not represent the complexity of actual patient encounters. Real-world patients frequently present with incomplete histories, multiple simultaneous conditions, atypical symptoms, and clinically relevant comorbidities.

Fourth, the evaluation measured treatment recommendation accuracy rather than patient outcomes. The study therefore cannot determine whether use of the software would improve morbidity, mortality, time to treatment, or other clinical outcomes.

Fifth, the grading criteria and question construction may influence the measured accuracy. Independent replication using externally generated cases would provide stronger evidence of generalizability.

Finally, the results do not establish that the software should be used autonomously to diagnose or treat patients. Clinical decision-support systems should be evaluated within appropriate clinical, regulatory, and professional frameworks.

6. Recommendations for Future Research

Future evaluations should expand the benchmark substantially and include a larger number of clinical conditions and patient presentations.

Particularly valuable next steps would include:

- Independent external validation using questions not used during system development.
- Evaluation on several hundred or thousands of clinical cases.
- Inclusion of atypical and ambiguous presentations.
- Evaluation of pediatric and geriatric cases.
- Testing cases involving multiple simultaneous diagnoses.
- Evaluation of medication contraindications and clinically important comorbidities.
- Comparison with practicing physicians, paramedics, nurses, or other appropriate clinicians.
- Assessment of inter-rater agreement among clinical evaluators.
- Prospective evaluation in simulated clinical environments.
- Measurement of sensitivity, specificity, positive predictive value, and negative predictive value where the study design supports these measures.
- Evaluation of treatment omissions, harmful recommendations, and inappropriate treatment escalation separately from simple correctness.
- Assessment of whether the system appropriately recognizes when a case requires emergency escalation or human clinical judgment.

These studies would provide substantially stronger evidence regarding the system's generalizability and clinical utility.

7. Conclusion

In this 50-question benchmark evaluation, the assessed AI-assisted clinical decision-support software achieved **100% treatment recommendation accuracy**, with 50 of 50 evaluated questions receiving recommendations that satisfied the predetermined grading criteria.

The system achieved 100% observed accuracy in each of the eight evaluated clinical categories: bronchospasm/asthma, spontaneous pneumothorax, respiratory failure, impending cardiac arrest, severe allergic reaction/anaphylaxis, appendicitis, gastrointestinal bleeding, and gallstones/cholecystitis.

These findings demonstrate strong performance on the evaluated benchmark. Nevertheless, a perfect score on a limited question set should be interpreted as an initial validation result rather than evidence of universal clinical accuracy or safety. Independent, larger-scale, clinically representative validation is necessary before conclusions can be made regarding real-world clinical effectiveness or autonomous clinical use.

8. Data Availability

The underlying benchmark questions, scoring criteria, and complete software outputs should be made available to reviewers or readers where permitted. Providing the complete evaluation dataset would facilitate independent replication of the reported findings.

9. Conflict of Interest

The authors should disclose any financial, employment, ownership, development, or other relationship with the software evaluated in this study.

10. Ethics Statement

Because this evaluation involved a benchmark of clinical questions rather than direct intervention with human patients, formal human-subject research requirements may not apply. The applicable institutional and regulatory requirements should nevertheless be reviewed and reported by the investigators.

11. References

The final manuscript should include authoritative clinical references supporting the diagnostic features and treatment standards used to construct and grade the benchmark. Educational websites or question-bank materials may be useful for identifying source material, but treatment recommendations in a scientific manuscript should preferably be supported by current professional guidelines, peer-reviewed literature, and authoritative clinical references.
